Data Science

Lecture 05: Neural Networks

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Where We Left Off

Lecture 04 fixed the form — a line, a log-odds line — and chose the features by hand. Today relaxes both: stack simple units until the form is flexible, and let the network build its own features.

Lecture 04
a fixed form, hand-made features

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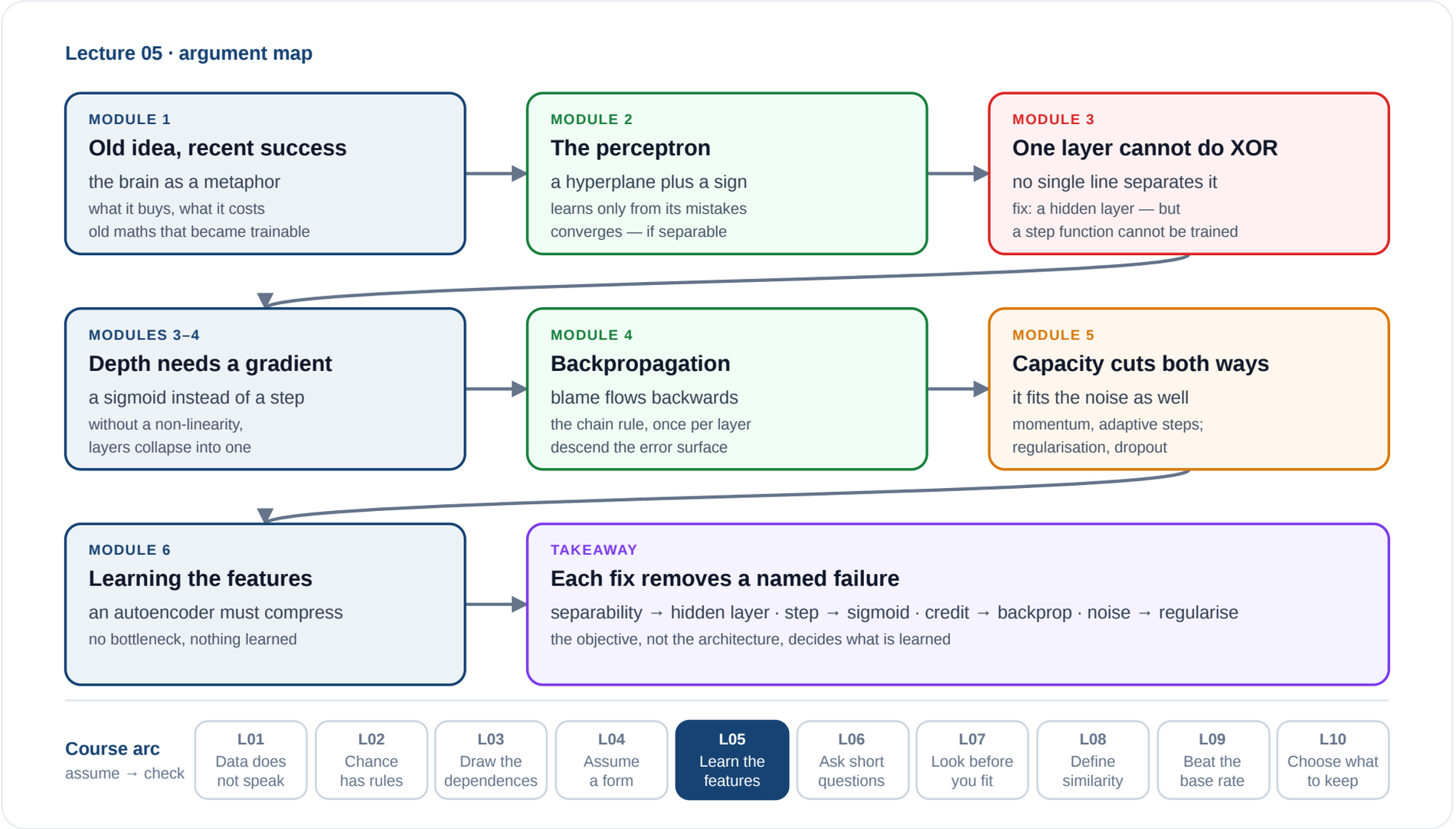
Today
a learned form, learned features

→

The price
capacity that also fits noise

> The perceptron is Lecture 04's linear classifier, the delta rule is LMS, and the sigmoid is logistic regression's — each returns with a new job.

Argument Map



Module 1

Where the Metaphor Comes From

> ***Guiding Question:*** *The perceptron dates from 1958. What does the brain metaphor buy — and why did the idea start to work only around 2012?*

The Neuron

≈ 86 billion

neurons in the human brain (estimated, not counted)

10^{14}

synapses (estimated)

10^{3}–10^{4}

synapses on a typical neuron — about 10^{14} / 10^{11}

10^{-3} **s**

the fastest a neuron switches

A neuron receives signals through its dendrites, integrates them, and fires along its axon when the combined input passes a threshold.

What the Metaphor Buys — and What It Costs

Worth borrowing

Many simple units, massively connected; behaviour that emerges from weights rather than from rules; learning by adjusting connection strengths.

Not worth claiming

Artificial neurons are **not** models of biological ones. Backpropagation has no established biological counterpart, and no one is simulating a brain.

"Neural network" is a historical name, not a claim about neuroscience. Treat brain analogies in marketing material accordingly.

Takeaway

> The brain is the metaphor not for its speed — each neuron is a million times slower than a transistor — but for its **architecture**: many simple units, massively parallel, whose behaviour is learned by adjusting connection strengths. That is what artificial networks borrow, and all they borrow.

The Idea Is Old; The Success Is Recent

The perceptron dates from 1958, backpropagation from the 1980s. The methods in this lecture are decades old. **What changed in 2012?**

More data

Large labelled datasets that did not previously exist (ImageNet, 2009).

More compute

GPUs able to run many passes over those datasets.

Better practice

Fixes available by 2012: ReLU activations, Glorot initialisation, dropout.

Followed the breakthrough rather than enabling it: ADAM (2014), batch normalisation and residual networks (both 2015).

What This Implies

> The bottleneck was never the idea. It was **data, compute and a set of practical fixes** — which is why "we tried neural networks in 1995 and they did not work" was both true and misleading.

Mini-Exercise

A colleague says: "Deep learning was invented in 2012." What is wrong with the claim, and what did 2012 bring instead?

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> The methods are decades old — perceptron 1958, backpropagation in the 1980s. 2012 brought the ability to **train them at scale**: data, GPUs and a few practical fixes came together.

Takeaway

> Neural networks are not new mathematics; they are old mathematics that became trainable. Every "new" technique below exists because a specific failure mode had to be removed.

Module 2

The Perceptron

> ***Guiding Question:*** *Least squares was nudged by examples it already classified correctly. What would a learning rule for decisions look like?*

The Model

A perceptron computes a weighted sum of its inputs and applies a **threshold**:

$$h_{\mathbf{w}}(\mathbf{x}) = \text{sign}\left(w_0 + \sum_{j=1}^{p} w_j x_j\right) = \text{sign}(\mathbf{w}^\top \mathbf{x})$$

Inputs

$x_1, \ldots, x_p$, plus a constant $x_0 = 1$ carrying the bias w_0

Weights

$w_0, w_1, \ldots, w_p$ — **the hypothesis is exactly this vector**

Output

$+1$ or -1: which side of the boundary the input lies on

This is the same object as *linear regression for classification* from Lecture 04 — a hyperplane and a sign. What is new is how the weights are found.

Weight Adaptation

Perceptron learning rule

For each training example $(\mathbf{x}, y)$ with $y \in \{-1, +1\}$:

$$\Delta w_j = \eta\big(y - h_{\mathbf{w}}(\mathbf{x})\big)x_j$$

Weights change **only when the example is misclassified**; correctly classified examples leave the hypothesis untouched.

Contrast with least squares, which is nudged by *every* example including the ones it already gets right. This is exactly the defect that made squared loss a poor classification loss.

The Perceptron Convergence Theorem

If the training data is **linearly separable**, the perceptron rule finds a separating hyperplane in a **finite** number of updates — regardless of the starting weights and of the learning rate.

What it promises

Termination with zero training error, guaranteed, when a solution exists.

What it does not promise

Anything at all when the data is **not** separable — the procedure simply never settles. And nothing about *which* separating hyperplane you get, or how it generalises.

Mini-Exercise

A perceptron with $w_0 = -1.5$ and $w_1 = w_2 = 1$ receives binary inputs. Which logical function does it compute — and which w_0 turns it into OR?

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> Sums $-1.5, -0.5, -0.5, +0.5$ for $(0,0), (0,1), (1,0), (1,1)$: only $(1,1)$ fires, so it is **AND**. With $w_0 = -0.5$ any single 1 clears the threshold: **OR**. The bias *is* the threshold, with its sign flipped.

Takeaway

> A perceptron is a **hyperplane plus a sign**, learned by correcting only its mistakes. Convergence is guaranteed — but only when a separating hyperplane exists at all.

Three Open Questions

Which solution?

Every separating hyperplane satisfies the theorem. The one you get depends on the initial weights and the order of the examples.

How long?

"Finite" can be very large. The bound grows as the **margin** shrinks — the distance from the separating hyperplane to the nearest training point.

And if not separable?

The procedure oscillates forever. Real data usually is not separable.

The theorem guarantees *a* separator on separable data — not which one, not how fast, not what happens otherwise. Zero training error is not the objective: a hyperplane with a tiny margin grazes the training points and may sit in the worst place for new ones.

Module 3

The Limit of One Layer

> ***Guiding Question:*** *You built AND and OR with a single perceptron. Why can you not build XOR — and what does that tell us about model families?*

The Question

You built AND and OR with a single perceptron. **Now build XOR:** output 1 when exactly one input is 1.

x_1	x_2	XOR
0	0	0
0	1	**1**
1	0	**1**
1	1	0

Try to find weights w_0, w_1, w_2. Take a minute.

Why It Cannot Work

Plot the four points. The two positive cases sit on **opposite corners** of the square, and so do the two negative ones.
A perceptron can only draw **one straight line**. No single line puts $(0,1)$ and $(1,0)$ on one side while keeping $(0,0)$ and $(1,1)$ on the other.
XOR is **not linearly separable** — and no amount of training will change that.

> The limit is in the **hypothesis space**, not in the learning rule.

The Consequence, Historically

Minsky and Papert's 1969 analysis of exactly this limitation is widely credited with the collapse of neural-network funding for over a decade.

> The criticism was mathematically correct and practically overdrawn: the fix — stack a second layer — was already known. What was missing was a way to **train** it.

The Fix: A Hidden Layer

XOR **is** expressible with two layers — for instance as $(x_1 \vee x_2) \wedge \neg(x_1 \wedge x_2)$: two perceptrons computing OR and NAND, and a third computing AND of their outputs.

> The hidden layer builds a **new representation** in which the problem *is* linearly separable. That is what depth buys: not more lines, but better coordinates to draw them in.

> A single perceptron is limited by what one hyperplane can separate. Stacking layers removes that limit — which is why the next question is how to train a stack at all.

Why the Threshold Has to Go

The step function

$\text{sign}(\mathbf{w}^\top \mathbf{x})$ has derivative **zero everywhere** it is defined, and no derivative at all at the boundary.

The sigmoid

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

smooth, bounded, and with a derivative that is non-zero where it matters.

Without a usable gradient there is no way to assign blame to a weight buried in a hidden layer. **Differentiability is what makes depth trainable.**

A Useful Property

$$\sigma'(z) = \sigma(z)\big(1 - \sigma(z)\big)$$

The derivative is expressible in terms of the output — so a neuron that has already computed its activation can compute its own gradient for free.

> Both tails are flat: for large $|z|$ the derivative is nearly zero and learning stalls. This is the **vanishing gradient** problem, and it is why ReLU replaced the sigmoid in deep networks.

Mini-Exercise

$\sigma(0) = 0.5$ and $\sigma(12) \approx 0.99999$. Using $\sigma' = \sigma(1 - \sigma)$: how large is the gradient at $z = 0$, and at $z = 12$?

Mini-Exercise

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> $\sigma'(0) = 0.5 \times 0.5 = 0.25$, the steepest the sigmoid ever gets. $\sigma'(12) \approx 0.00001$: the update is effectively zero. The neuron is **saturated** and stops learning.

Takeaway

> The extra layer makes XOR *expressible*; the sigmoid makes the stack *trainable* — differentiability is the precondition for training depth. The price is saturating tails where learning stalls.

Module 4

Depth, and How to Train It

> ***Guiding Question:*** *A hidden neuron has no target of its own. How much of the final error is it responsible for?*

Structure

Layer

$L_i = \{n_{1,i}, \ldots, n_{k,i}\}$ — a set of neurons. L_0 is the input layer.

Fully connected

Every neuron in L_{i-1} feeds every neuron in L_i.

Parameters

$\mathbf{W}_i \in \mathbb{R}^{|L_i| \times |L_{i-1}|}$ — one row of weights per neuron — and a bias vector $\mathbf{b}_i$, one bias per neuron.

The forward pass is then a chain of matrix products with a non-linearity between them:

$$\mathbf{a}_i = \sigma\big(\mathbf{W}_i \mathbf{a}_{i-1} + \mathbf{b}_i\big), \qquad \mathbf{a}_0 = \mathbf{x}$$

Why the Non-Linearity Is Not Optional

Without σ, a stack of layers is $\mathbf{W}_3\mathbf{W}_2\mathbf{W}_1\mathbf{x}$ (plus biases) — a **single matrix**, hence a single linear model. Depth without non-linearity buys nothing at all.

Each layer transforms the representation; the final layer draws a hyperplane in the coordinates the earlier layers built. This is XOR's fix, generalised.

Mini-Exercise

A network has 4 inputs, one hidden layer of 5 neurons and 3 outputs, fully connected, with a bias per neuron. How many parameters does it have?



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$$\underbrace{(4 \times 5) + 5}_{\text{hidden layer}} + \underbrace{(5 \times 3) + 3}_{\text{output layer}} = 25 + 18 = 43$$

> Counts grow with the **product** of adjacent layer sizes, which is why parameter counts explode with width.

Takeaway

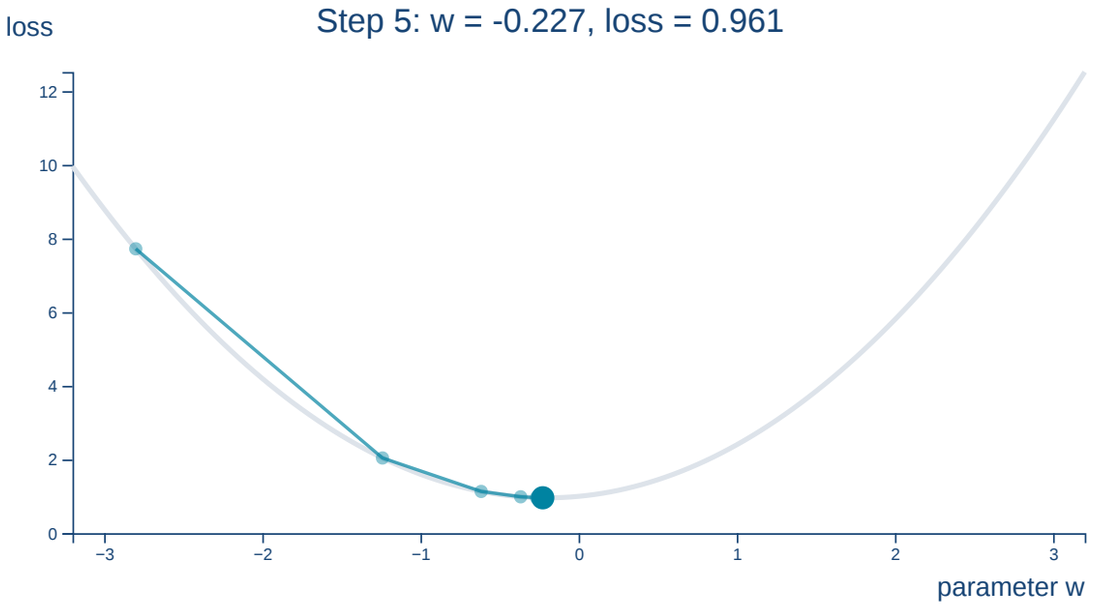
A multilayer network is matrix products separated by non-linearities. **The non-linearity is what makes depth mean anything**, and each hidden layer is a learned change of coordinates.

The Error Surface

For weights $\mathbf{w}$ and training set D, the squared error

$$E(\mathbf{w}) = \frac{1}{2} \sum_{(\mathbf{x},y)\in D} \big(y - h_{\mathbf{w}}(\mathbf{x})\big)^2$$

is a **surface over the weight space**. Training means walking downhill on it.



Two Ways to Descend

Batch gradient descent

Accumulate the gradient over **all** examples, then take one step.

Stable, expensive per step, and it needs the whole dataset in memory.

Incremental (stochastic) gradient descent

Update after **each** example.

Noisier, far cheaper per step, and the noise can help escape poor regions.

The **delta rule**, $\Delta w_j = \eta(y - \mathbf{w}^\top \mathbf{x})x_j$, is incremental gradient descent on the squared error — the LMS rule from Lecture 04 under a new name.

Mini-Exercise

You train on 1,000,000 examples, and one pass over them takes an hour. How many weight updates per hour does **batch** gradient descent make — and how many does **incremental**?

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You train on 1,000,000 examples, and one pass over them takes an hour. How many weight updates per hour does **batch** gradient descent make — and how many does **incremental**?

> Batch: **one**. Incremental: **one million**, each on a noisy estimate of the gradient. Progress per unit of computation is vastly higher — at the risk of instability if η is too large.

Takeaway

Training is descent on an error surface. **Batch** steps are accurate and rare; **incremental** steps are noisy and frequent — and on real data, frequency wins.

The Credit Assignment Problem

The output layer's error is measurable: we know the target. But a hidden neuron has **no target of its own**.

> *How much of the final error is this hidden weight responsible for?*

The Answer: Blame Flows Backwards

Output layer

The error is measured against the target, times the slope of the activation: $\delta_o = (y - o)\sigma'(z_o)$.

Hidden layers

The error of a neuron is the **weighted sum of the errors of the neurons it feeds**, times its own slope: $\delta_h = \sigma'(z_h) \sum_k w_{hk}\delta_k$.

A neuron that contributed strongly to a wrong answer receives a large share of the blame. That is the whole idea; the calculus is the chain rule applied layer by layer.

The Algorithm

Backpropagation (incremental form)

repeat **for each** $(\mathbf{x}, y) \in D$: **forward:** compute activations layer by layer up to the output **backward:** compute the output error, then propagate it to each earlier layer **update:** $w \leftarrow w + \eta \cdot \delta_{\text{target}} \cdot a_{\text{source}}$ **until** convergence **or** $t > t_{\max}$

The backward pass costs about **twice** the forward pass, so one training step costs about **three** times inference — the fact behind every training-cost estimate you read.

Mini-Exercise

A hidden neuron feeds two outputs through weights 0.8 and -0.2; their errors are $\delta_1 = 0.5$ and $\delta_2 = 1.0$. What is its propagated error, before the σ' factor?

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> $0.8 \times 0.5 + (-0.2) \times 1.0 = 0.4 - 0.2 = 0.2$. Blame is weighted by **weight × error**: output 1 contributes more, and the negative weight partly cancels it.

Takeaway

Backpropagation solves **credit assignment**: error flows backwards, weighted by the connections that carried the signal forward and scaled by each neuron's slope σ'. It is the chain rule, applied once per layer — no more and no less.

Module 5

Making Training Work

> ***Guiding Question:*** *The network fits the training data perfectly and fails on new data. Which of your assumptions was wrong?*

Two Failure Modes

Local minima

A point lowest in its neighbourhood, $E(\mathbf{w}_{\text{local}}) \leq E(\mathbf{w})$ for all $\mathbf{w}$ near it — yet not globally: $E(\mathbf{w}_{\text{opt}}) < E(\mathbf{w}_{\text{local}})$.

In practice, large networks appear to have **many** local minima, most of them close in value to the best one — so this matters less than intuition suggests.

Flat regions

Near-zero gradients produce near-zero steps. Training appears to have converged when it has merely stalled — the sigmoid's saturated tails are the usual cause.

Momentum

$$\Delta \mathbf{w}_t = -\eta \nabla E(\mathbf{w}_t) + \alpha \Delta \mathbf{w}_{t-1}$$

Each step keeps a fraction α of the previous one: consistent directions accumulate speed, oscillating directions cancel.

> Momentum helps through flat regions and damps zig-zagging in narrow valleys — but the learning rate η remains critical for convergence.

Adaptive Learning Rates

AdaGrad

One learning rate **per parameter**: η divided by the square root of that parameter's accumulated squared gradients.

Parameters with large gradients slow down quickly; rarely-updated parameters keep a large step size.

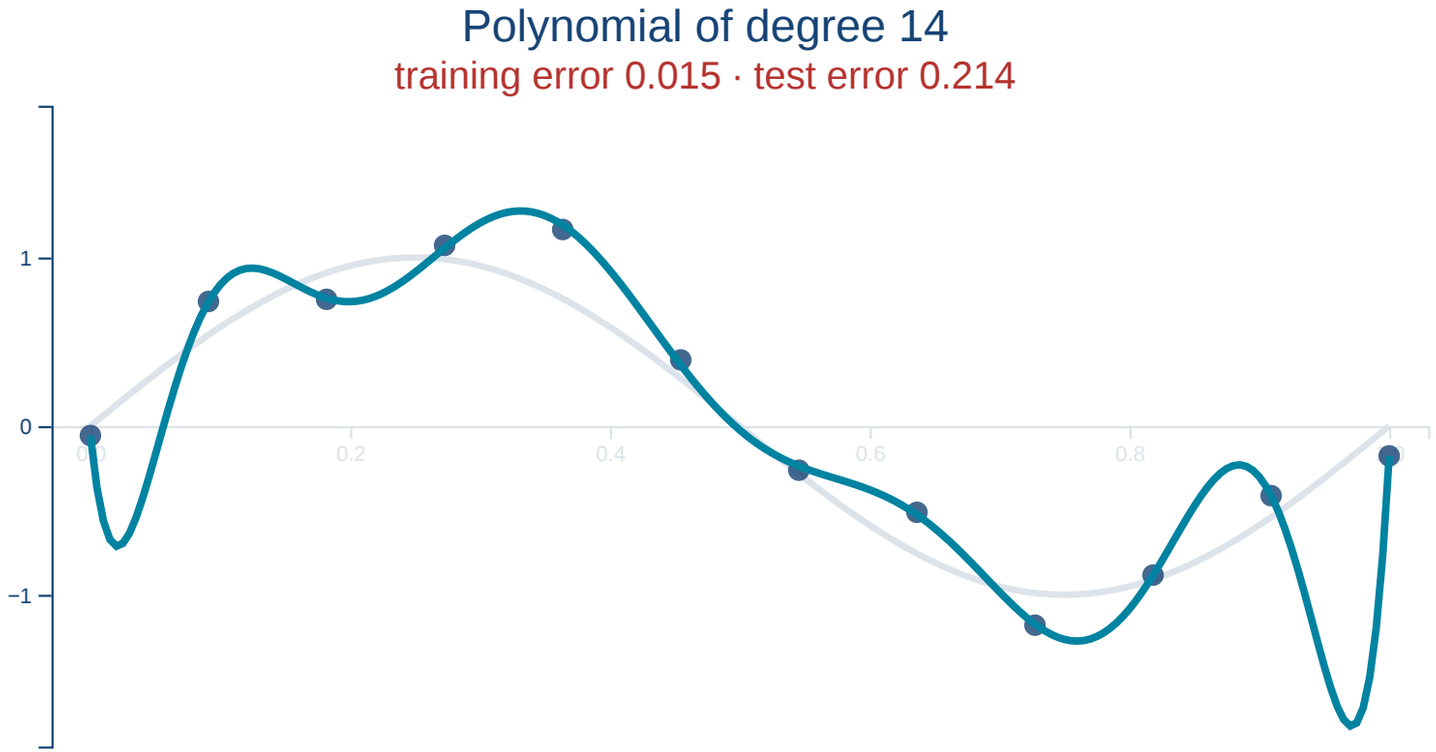
And its successors

RMSProp replaces the ever-growing sum with a moving average; **ADAM** combines that with momentum, and is the common default today.

The obstacles are flat regions and badly scaled parameters, not mythical local minima. **Momentum** adds inertia, **adaptive methods** give each parameter its own step size — and both exist to remove a specific, diagnosable failure.

Capacity Cuts Both Ways

A network with enough neurons can fit **any** training set — including its noise. The decision surface becomes intricate where the data is sparse.



Regularization

Add a penalty on the weights to the loss:

$$E_{\text{reg}}(\mathbf{w}) = E(\mathbf{w}) + \lambda\Omega(\mathbf{w})$$

where λ controls how much the penalty counts.

L_2 — $\Omega = \|\mathbf{w}\|_2^2 = \sum_j w_j^2$

Penalises **peaky** weight vectors and prefers diffuse ones: every input gets used a little. The default choice.

L_1 — $\Omega = \|\mathbf{w}\|_1 = \sum_j |w_j|$

Produces **sparse** weights (many exact zeros): a neuron keeps only its most important inputs — a feature selector.

Dropout

During training, each neuron's output is **silenced with probability** $1 - p$. The network cannot rely on any single unit, so it must spread the representation across many.
Because training silences a fraction of the units, a layer's total output is smaller during training than at inference. The surviving activations are therefore scaled by $\frac{1}{p}$ to compensate.

Dropout is active during **training only**. Forgetting to disable it at inference time is a classic bug.

Mini-Exercise

Your network reaches 99% training accuracy and 71% validation accuracy. What is the diagnosis, and which intervention would you try first?

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> **Overfitting** — a 28-point gap. More data if obtainable: it reduces variance without adding assumptions. Otherwise stronger regularisation (λ, dropout), a smaller network or early stopping — cheap to try, and the gap shows whether it worked.

Takeaway

Capacity that can fit the signal can also fit the noise. **Regularisation is inductive bias made explicit**: it encodes what you believe about the unseen data — and the gap between training and validation error is how you find out whether you believed correctly.

Module 6

Learning the Representation

> ***Guiding Question:*** *In Lecture 04 we chose the features by hand. Can a network choose them for us — and how would we know if it did?*

Learning to Reproduce the Input

An **autoencoder** is trained so that its output reproduces its input, through a hidden layer that is **narrower** than the input.

Encoder

maps the input to the narrow hidden layer

Bottleneck

a compressed code — the representation being learned

Decoder

reconstructs the input from that code

No labels are required: the target *is* the input.

What Makes It Work

The bottleneck does the work

A narrow hidden layer is a **lossy compression** of the input: the network must discard whatever it can afford to lose.

Too wide and it cheats

If the hidden layer is large enough, the network simply learns the **identity function** and has represented nothing.

Two Refinements

Denoising autoencoders

Corrupt the input, then require reconstruction of the *clean* version — the identity function is no longer a solution.

Weight sharing

Tie the decoder to the encoder, $\mathbf{W}_{1,2} = \mathbf{W}_{0,1}^{\top}$, halving the weights to learn.

For binary inputs, cross-entropy replaces squared error as the reconstruction loss — the same substitution as in logistic regression.

How Would We Know It Worked?

A low training reconstruction error proves little — the identity function has none. Two checks do:

Held-out reconstruction

Reconstruction error on data the network never saw: the code must generalise, not memorise.

Downstream use

Train a simple model on the code, with few labels. If it beats the same model on the raw inputs, the representation has captured structure.

> A learned representation is judged like any model: on unseen data, and by what it lets us do — not by how well it fits its own training set.

Mini-Exercise

An autoencoder with 784 inputs and a 784-neuron hidden layer reaches almost zero reconstruction error. Is this a good result?

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An autoencoder with 784 inputs and a 784-neuron hidden layer reaches almost zero reconstruction error. Is this a good result?

> No — a warning sign. With no bottleneck it has probably learned the **identity**: copy the input across, represent nothing. Narrow the bottleneck, or corrupt the input (denoising), so copying no longer solves the task.

Takeaway

> An autoencoder learns a representation by being **forced to lose information**, and is judged on unseen data and downstream use. Remove the constraint and it learns nothing — the objective, not the architecture, decides what a model learns.